Optimizing Chunk Size and Overlap for FAISS Retrieval on NPC Complaint Procedures

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***Abstract*—This study investigates how fixed-size text chunking and overlap affect retrieval performance for National Privacy Commission (NPC) complaint procedures. Twelve configurations were evaluated on pages 10–17 of a 35-page compiled PDF using the same sentence-embedding model, FAISS inner-product search, and nine questions from three difficulty levels. Each test retrieved the top three passages, and performance was measured using Precision@3, Recall@3, and mean reciprocal rank at three (MRR@3). In the automatically evaluated Google Colab experiment, C12 (1,500-character size and 200-character overlap) achieved Precision@3 = 0.5926, Recall@3 = 0.9167, and MRR@3 = 0.8148 while generating 23 chunks. C9 achieved the highest Precision@3 (0.6296), and C2 achieved the highest MRR@3 (0.8889). C12 was recommended provisionally for its strong overall evidence coverage and especially its difficult-question recall (0.9167). The relevance labels were largely based on fixed phrase matching, so independent review of the retrieved passages remains necessary.**

Keywords—document chunking, FAISS, retrieval evaluation, National Privacy Commission, Precision@3, Recall@3, MRR@3.

I. Introduction

Retrieval-augmented generation (RAG) systems [6] first search a document collection for passages relevant to a question, then may use the retrieved content to produce a grounded response. When source material is divided into fixed-size chunks, chunk size and overlap determine what text can be retrieved together. Small chunks may focus on a single fact but lose surrounding context; larger chunks may preserve related information but also contain extra material. Overlap may reduce boundary splits, although it increases redundant text and the number of stored vectors [3]–[5].

This experiment focuses on retrieval, not generated-answer accuracy. It measures how much supporting evidence FAISS returns from an NPC procedural corpus under controlled text-splitting settings. The goal is to choose a configuration that covers evidence for multi-part filing questions while keeping indexing and searching practical.

II. Dataset Description

A. Dataset and document scope

The research dataset, NPC_Dataset.pdf, is a 35-page compiled preparation corpus on complaint procedures of the Philippine National Privacy Commission. It summarizes public source documents including the 2025 Citizen's Charter and the NPC Rules of Procedure as amended in 2024 [1], [2]. The entire PDF was retained as the source artifact, but this controlled Part 2 experiment indexed only PDF pages 10–17, covering filing forms, submission channels, fees and payment, docketing, filing modes, service times, and case assignment. These are PDF page numbers in the compiled dataset, not the page numbers of the original NPC publications.

The corpus contains headings, narrative explanations, source-reference lines and occasional process-related distinctions. It is not a set of real complaints or private case records. This makes it suitable for testing whether retrieval can distinguish administrative intake steps from later legal proceedings.

B. Test questions and reference evidence

Nine predefined questions were used: Q1–Q3 are Easy, Q4–Q6 are Average, and Q7–Q9 are Difficult. Questions include single-document facts, related filing steps, and comparisons across sections. There are 31 predefined evidence units overall, with two to five units per question. Examples include the two payment routes (Q1), proof-of-payment and docketing steps (Q5), the distinction between inquiries and formal filing (Q7), and the difference between a one-hour filing-service estimate and a five-day officer-assignment period (Q9). The same questions and evidence definitions were used across all configurations.

III. EXPERIMENTAL METHODOLOGY

A. Configuration and baseline control

Twelve configurations (C1–C12) varied chunk size from 300 to 1,500 characters and overlap from 0 to 200 characters. The Part 1 baseline is C11 (1,000 characters, 200 overlap). C1–C5 follow the combinations named in the laboratory handout; the remaining settings extend comparisons at 0%, 10% and 20% overlap where possible. C12 uses 1,500 characters with 200 characters of overlap, about 13.3% overlap, so it is not treated as an exact same-percentage comparison against the other groups.

For every configuration, the program regenerated chunks and embeddings and built a fresh FAISS index. The source pages, question text, embedding model, vector normalization, search method, and top-k value were kept constant. The experiment used all-MiniLM-L6-v2 sentence embeddings (384 dimensions), normalized embeddings, FAISS inner-product similarity, and exactly k = 3 retrieved chunks [3], [4]. GPT-OSS 20B was retained as the optional Part 1 answer-generation model, but no language-model generation was required for these retrieval metrics.

B. Evaluation measures and relevance labels

Precision@3 is the number of relevant passages among the top three divided by three. Recall@3 is the number of distinct required evidence units covered by those passages divided by the total required for the question. Reciprocal rank is 1/r when the first relevant chunk appears at rank $r \leq 3$, otherwise zero. MRR@3 is the arithmetic mean of

reciprocal ranks across all nine questions. The same rule is also applied separately to the three easy, three average, and three difficult questions [5].

The notebook first labels retrieved passages using a consistent set of accepted phrases for each predefined evidence unit. This creates repeatable automated judgments, but a matching phrase may not always supply enough context, while a relevant paraphrase may be missed. The results in this report are the original automatically evaluated metrics from the submitted Colab Excel workbook and original detailed retrieval log. A separate later review identified eight possible label corrections, but those adjusted metrics are deliberately not mixed into the results reported here. Consequently, these scores must be interpreted as provisional until all 324 retrieved passages are checked against the source PDF.

C. *Recorded efficiency measures*

For each configuration, the experiment recorded the total number of chunks, embedding time, FAISS index-building time, average retrieval latency per question, and a vector-storage estimate. The estimate is based on the number of 384-dimensional vectors, not a measured serialized FAISS file size. Timings reflect one Colab run and are not a statistical benchmark of speed.

IV. Results

A. *Overall configuration comparison*

Table I gives the configuration-level means calculated from the same nine questions for all twelve settings, using the original exported Colab results. C9 achieved the highest mean Precision@3 (0.6296), C12 achieved the highest Recall@3 (0.9167), and C2 achieved the highest MRR@3 (0.8889). The winners differ across the metrics. C12 has 23 chunks and balances evidence coverage with a comparatively small vector index, even though it does not lead on precision or reciprocal rank.

TABLE I. CONFIGURATION-LEVEL RETRIEVAL RESULTS (ORIGINAL AUTOMATED SCORES)

Cfg	Size / Ov.	Chunks	P@3	R@3	MRR@3
C1	300/0	94	0.4074	0.6352	0.7037
C2	300/30	107	0.4444	0.5574	0.8889
C3	300/60	117	0.3704	0.5389	0.5926
C4	500/0	59	0.4815	0.5611	0.7222
C5	500/50	65	0.4815	0.6907	0.7593
C6	500/100	73	0.5185	0.7222	0.7778
C7	750/0	41	0.4815	0.7500	0.7037
C8	750/75	44	0.5556	0.7278	0.8333
C9	1000/0	30	0.6296	0.8056	0.8333
C10	1000/100	35	0.5556	0.8056	0.8333
C11	1000/200	38	0.5556	0.6944	0.7778
C12	1500/200	23	0.5926	0.9167	0.8148

B. *Metrics across the twelve settings*

Fig. 1 compares the original Precision@3, Recall@3, and MRR@3 scores for C1–C12. Increasing chunk size generally improved evidence coverage across several tested settings, but neither chunk size nor overlap gave consistent improvements in every metric. The chart uses the same values saved in the submitted Excel workbook; it does not show the later partially adjusted audit results.

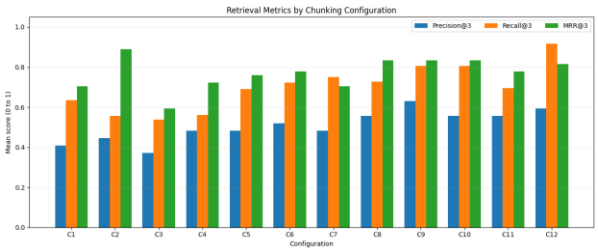


Fig. 1. Original Precision@3, Recall@3 and MRR@3 for C1-C12.

C. *Performance by question difficulty*

Table II presents performance for the three Easy, three Average, and three Difficult questions under each configuration (36 configuration-by-difficulty rows). C12 achieved difficult-question Recall@3 of 0.9167, compared with 0.4167 for baseline C11 and 0.5833 for C9. The breakdown matters because a strong overall average can hide incomplete evidence coverage on multi-part questions.

TABLE II. PERFORMANCE BY DIFFICULTY (ORIGINAL MEANS)

Cfg	Level	P@3	R@3	MRR@3
C1	E	0.3333	0.7222	0.6111
C1	A	0.5556	0.6833	1.0000
C1	D	0.3333	0.5000	0.5000
C2	E	0.2222	0.5556	0.6667
C2	A	0.6667	0.7000	1.0000
C2	D	0.4444	0.4167	1.0000
C3	E	0.2222	0.6667	0.2778
C3	A	0.4444	0.5333	1.0000
C3	D	0.4444	0.4167	0.5000
C4	E	0.4444	0.6667	0.5000
C4	A	0.5556	0.6000	0.6667
C4	D	0.4444	0.4167	1.0000
C5	E	0.2222	0.5556	0.4444
C5	A	0.5556	0.9333	0.8333
C5	D	0.6667	0.5833	1.0000
C6	E	0.3333	0.6667	0.6667
C6	A	0.6667	1.0000	0.8333
C6	D	0.5556	0.5000	0.8333
C7	E	0.3333	0.6667	0.6667
C7	A	0.6667	0.9167	1.0000
C7	D	0.4444	0.6667	0.4444
C8	E	0.4444	0.6667	0.6667
C8	A	0.6667	0.8500	1.0000
C8	D	0.5556	0.6667	0.8333
C9	E	0.5556	0.8333	0.6667
C9	A	0.6667	1.0000	1.0000
C9	D	0.6667	0.5833	0.8333
C10	E	0.5556	0.6667	0.6667
C10	A	0.5556	1.0000	1.0000
C10	D	0.5556	0.7500	0.8333

Cfg	Level	P@3	R@3	MRR@3
C11	E	0.5556	0.6667	0.6667
C11	A	0.6667	1.0000	1.0000
C11	D	0.4444	0.4167	0.6667
C12	E	0.6667	0.8333	0.7778
C12	A	0.5556	1.0000	0.8333
C12	D	0.5556	0.9167	0.8333

V. Discussion

A. Effects of chunk size and overlap

At zero overlap, C1 (300/0) creates 94 chunks and obtains Recall@3 = 0.6352; C7 (750/0) creates 41 chunks with recall 0.7500; and C9 (1,000/0) creates 30 chunks with recall 0.8056. Larger zero-overlap chunks retained more of the needed evidence within this specific range, although the trend cannot be assumed to extend to all datasets.

Holding chunk size at 1,000 characters provides an overlap-controlled comparison. C9 with zero overlap produced 30 chunks and Recall@3 = 0.8056; C10 with 100-character overlap produced 35 chunks and the same Recall@3 = 0.8056; and baseline C11 with 200-character overlap produced 38 chunks but lower Recall@3 = 0.6944. More overlap therefore added stored chunks without improving recall in this specific comparison. The effect depends on where supporting text falls relative to chunk boundaries.

B. Successful and incomplete retrieval examples

C12 recovered all five predefined evidence units for Q5, concerning proof of payment, validation, docketing, and the returned receipt, under the automatic matching criteria. It also recorded Recall@3 = 1.0000 for Q8 on filing modes and Q9 on the differences between the one-hour filing-service estimate and the five-calendar-day case-assignment period. However, its Q7 recall was only 0.7500, indicating that one of four units was not detected by the phrase matcher. These are examples of measured retrieval coverage, not proof of legal accuracy.

Not every retrieval was complete. C12 found only one of the two required evidence units for Q1 (Recall@3 = 0.5000), so the payment-route question was incompletely covered in its top three. C9 retrieved only one of four evidence units for Q9 (Recall@3 = 0.2500), despite relatively strong overall Precision@3. These cases illustrate why a single high metric does not guarantee complete answers to multi-part questions.

C. Efficiency and consistency trade-offs

The Part 1 baseline C11 generated 38 chunks, whereas C12 generated 23 (39.5% fewer). Their estimated raw embedding-vector storage was 57.0 KB and 34.5 KB, respectively. In the latest Excel export, embedding times were 0.0985 s (C11) and 0.1057 s (C12); average retrieval latencies were 19.23 ms (C11) and 11.71 ms (C12). These are single-run timings and should not be used to claim a statistically reliable speed improvement. The storage estimate is the raw vector footprint, not a measured FAISS file size.

C12's Recall@3 = 0.9167 was higher than C9's 0.8056 and C11's 0.6944, and C12 retrieved 29 of the 31 predefined evidence units across the nine questions according to the saved labels. Eight questions reached full evidence-unit coverage, while Q1 remained incomplete. On Difficult questions alone, its mean Recall@3 was 0.9167, compared with 0.5833 for C9 and 0.4167 for baseline C11. Its Precision@3 = 0.5926 remained below C9's 0.6296, and its MRR@3 = 0.8148 remained below C2's 0.8889. These trade-offs favor C12 for multi-step evidence coverage rather than universal metric superiority.

D. Limitations and reliability

The experiment tests only nine questions from a selected eight-page excerpt of a compiled source PDF, so the findings cannot establish a general optimum for other corpora. Automated evidence judgments rely on accepted source phrases and may misclassify contextual support or paraphrases. Although a separate later audit identified eight cases requiring reconsideration, the original Excel and this paper consistently retain the notebook's automated labels. A complete manual audit of all 324 passages, a larger test-question set, and repeated timing trials would be needed to validate the results. The study measures retrieval, not the factual or legal correctness of generated answers.

VI. Final Recommendation

We provisionally recommend C12 (chunk size 1,500 characters, overlap 200 characters) for the selected NPC complaint-procedure pages. Its Recall@3 = 0.9167 is the highest of the twelve original experiment configurations; it also achieved Precision@3 = 0.5926 and MRR@3 = 0.8148. On Difficult questions, C12 attained Recall@3 = 0.9167 and generated only 23 chunks, compared with 38 for baseline C11. C9 offers higher precision and C2 higher MRR, so this recommendation gives greater weight to retrieving multiple required evidence units, not to a single leading metric.

This recommendation remains conditional. C12 retrieved only one of two payment-route evidence units in Q1 and achieved lower average MRR than C2. Its efficiency advantage is a smaller vector count; timings from one run do not establish a general speed advantage. The phrase-assisted relevance labels must also be manually checked against the NPC PDF before the reported ranking is treated as independently verified.

VII. Reproducibility

The reproducibility materials include the executed, documented Jupyter notebook, NPC_Dataset.pdf, the same nine questions and 31 predefined evidence units, configuration definitions, the original 324-row detailed retrieval log, and the Excel workbook with configuration results and difficulty averages. The notebook extracts PDF pages 10–17; for each (size, overlap) setting it regenerates chunks, encodes them with all-MiniLM-L6-v2, normalizes 384-dimensional embeddings, creates a new FAISS inner-product index, and retrieves the top three passages for Q1–Q9. The saved log supports calculation of the original metrics shown in this report.

For exact reproduction, use the same version of the source PDF, the same PDF page selection, embedding model

and normalization, nine test questions, and fixed evidence-matching rules. The report's tables and chart correspond to the original Config_Results, Difficulty_Results, and Detailed_Retrieval_Log worksheets in PTF2_Chunking_Experiment_Results.xlsx. Use the companion original retrieval log extracted from that workbook, not the later PTF2_Detailed_Log_8_Adjusted.csv, when submitting this version of the report. Any later evidence corrections should be accompanied by a separately recalculated workbook and revised report.

VIII. Conclusion

The tested chunking configurations differ in evidence coverage, ranking behavior, and number of stored vectors. Based on the original controlled Colab evaluation, C12 provides the best mean evidence recall and strong Difficult-question coverage with only 23 chunks. The preferred configuration is provisional because the ground-truth passage labels have not undergone complete independent manual review.

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